QUESTION 1

What does the term “Instruction Set Architecture (ISA)” refer to in a processor?

Answer is Wrong **Score: 0 / 1**

Options

A. The physical size of the chip

Expected

B. The set of instructions a processor can execute

C. The memory type used

Wrong

D. The speed of the processor

FEEDBACK

Solution : ISA is the specific set of instructions a processor can understand and execute.

Youtube link: https://youtu.be/zyiJIgYrnqg?si=q7NeuZw1ONh2pls0&t=142

QUESTION 2

Which of the following operations are core tasks of a processor?

Not Attempted **Score: 0 / 1**

Options

Expected
A. Arithmetic and logic operations

Expected
B. Loading/storing data from memory

C. Managing Wi-Fi connections

Expected
D. Control flow like branch/jump

FEEDBACK

Solution : Arithmetic/logic, memory load/store, and control flow as the processor's three main tasks.

Youtube link: https://youtu.be/zyiJIgYrnqg?si=q7NeuZw1ONh2pls0&t=142

QUESTION 3

What was a key feature that made AVR microcontrollers popular in early designs?

Not Attempted **Score: 0 / 1**

Options

A. Built-in Wi-Fi

Expected

B. Flash-based firmware upgradability

C. 64-bit registers

D. GPU integration

FEEDBACK

Solution : AVR's flash memory allowed easy firmware updates without complex programmers.

Youtube link: https://youtu.be/zyiJIgYrnqg?si=q7NeuZw1ONh2pls0&t=142

QUESTION 4

Why is an 8-bit processor considered less suitable for heavy computations?

Not Attempted **Score: 0 / 1**

Options

A. It overheats easily

Expected

B. It has limited address space and needs multiple cycles for big math

C. It lacks a power-saving mode

D. It cannot run any code

FEEDBACK

solution : The 8-bit processors have limited memory access and require more steps for large integer operations.

Youtube link : https://youtu.be/zyiJIgYrnqg?si=q0wxrf0bgWiedS-u

QUESTION 5

What is one reason Arduino is credited with democratizing MCU coding?

Not Attempted **Score: 0 / 1**

Options

A. It introduced new ISAs

Expected

B. It simplified setup with USB programming and an easy IDE

C. It removed the need for any code

D. It replaced all other microcontrollers

FEEDBACK

solution : Arduino's simplified hardware/software setup made MCUs accessible to everyone. Youtube link: https://youtu.be/zyiJIgYrnqg?si=SmbP2DGKcJCBenph&t=372

QUESTION 6

The PIC microcontroller originally used which type of memory for program storage?

Not Attempted **Score: 0 / 1**

Options

A. Flash memory

Expected

B. Electrically Erasable PROM (E-PROM)

C. ROM

D. DRAM

FEEDBACK

solution : PIC used E-PROM erasable with UV light through a small window. Youtube link: https://youtu.be/zyiJIgYrnqg?si=6b4Ug9dfWPn6ANra&t=454

QUESTION 7

What is a notable feature of the MSP430 series from Texas Instruments?

Not Attempted **Score: 0 / 1**

Options

A. High-power GPUs

Expected

B. Ultra-low power standby current

C. Built-in machine learning accelerators

D. Native 5G support

FEEDBACK

solution : MSP430’s extremely low standby currents, ideal for coin-cell applications. Youtube link : https://youtu.be/zyiJIgYrnqg?si=VkaiDBcyRv0gIXjP&t=557

QUESTION 8

Why are ESP32 microcontrollers widely popular?

Not Attempted **Score: 0 / 1**

Options

Expected

A. Very low cost

Expected

B. Integrated Bluetooth/Wi-Fi

C. Requires external debug boards only

D. Proprietary compilers only

FEEDBACK

solution : ESP32 offers low cost and builtin wireless connectivity, enabling quick IoT designs. Youtube link: https://youtu.be/zyiJIgYrnqg?si=6VoHA9UVWpT8Ws7b&t=701

QUESTION 9

How does ARM distribute its processor technology?

Not Attempted **Score: 0 / 1**

Options

A. It manufactures and sells chips

Expected

B. It licenses its processor designs to others

C. It only sells finished products

D. It provides free hardware to companies

FEEDBACK

solution : The ARM sells processor designs as intellectual property, not chips. Youtube link : https://youtu.be/zyiJIgYrnqg?si=mMuhFOh5bZ7P1Hzx&t=1003

QUESTION 10

In ARM Cortex architectures, which series is primarily targeted at microcontroller applications?

Not Attempted **Score: 0 / 1**

Options

A. Cortex-A

B. Cortex-R

Expected

C. Cortex-M

D. Cortex-X

FEEDBACK

solution : Cortex-M is designed for low-power embedded microcontroller applications.

Youtube link: https://youtu.be/zyiJIgYrnqg?si=vUM1sFoLtov7OcIz&t=1249